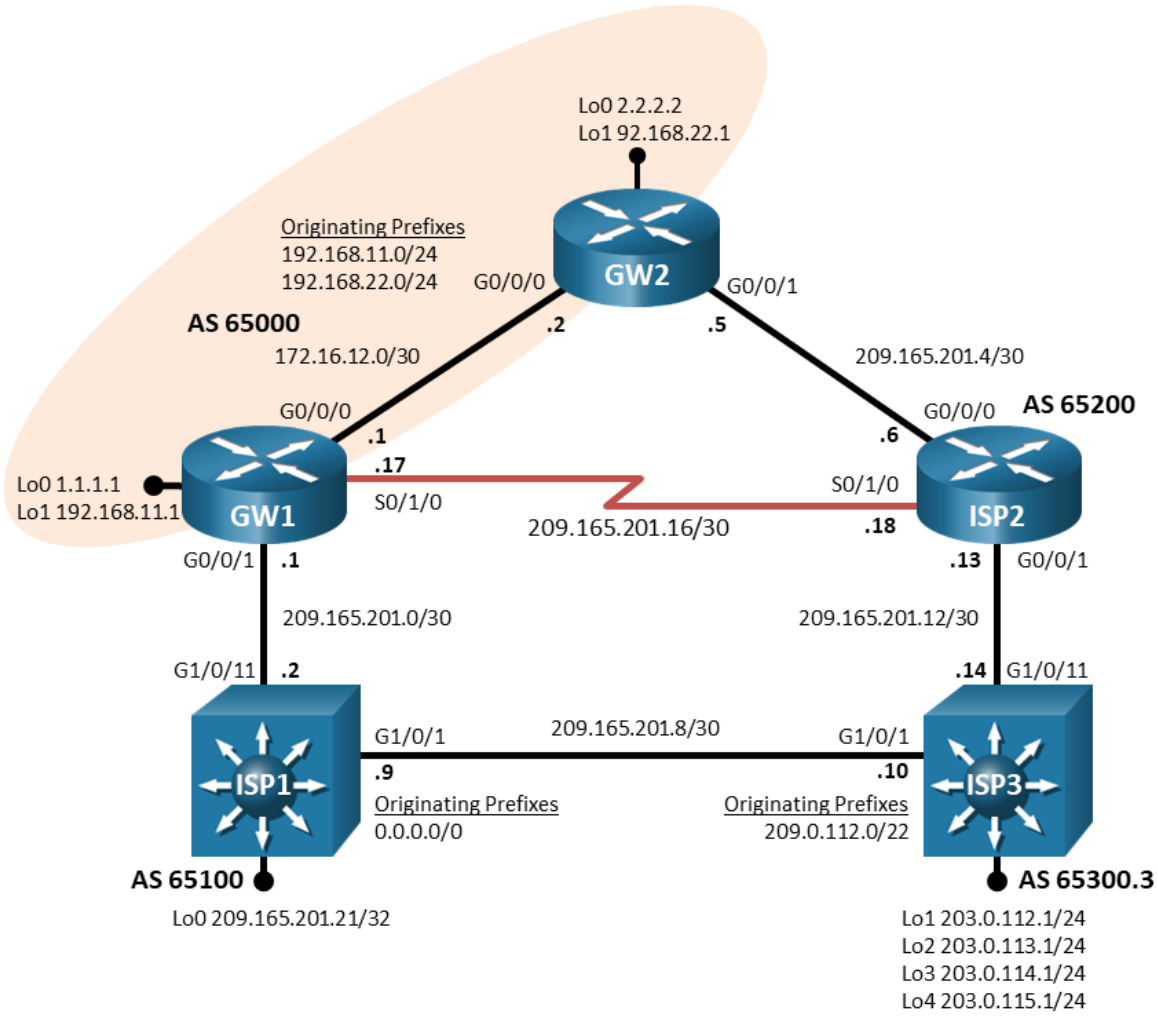
Lab - Configure Advanced BGP Features

Topology



Addressing Table

Device	Interface	IPv4 Address	IPv6 Link-Local
GW1	Loopback0	1.1.1.1/32	N/A
	Loopback1	192.168.11.1/24	N/A
	G0/0/0	172.16.12.1/30	EUI-64
	G0/0/1	209.165.201.1/30	N/A
	S0/1/0	209.165.201.17/30	N/A
GW2	Loopback0	2.2.2.2/32	N/A

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Device	Interface	IPv4 Address	IPv6 Link-Local
	Loopback1	192.168.22.1/24	N/A
	G0/0/0	172.16.12.2/30	EUI-64
	G0/0/1	209.165.201.5/30	N/A
ISP2	G0/0/0	209.165.201.6/30	N/A
	S0/1/0	209.165.201.18/30	N/A
	G0/0/1	209.165.201.13/30	N/A
ISP1	Loopback0	209.165.201.21/32	N/A
	G1/0/11	209.165.201.2/30	N/A
	G1/0/1	209.165.201.9/30	N/A
ISP3	Loopback1	203.0.112.1/24	N/A
	Loopback2	203.0.113.1/24	N/A
	Loopback3	203.0.114.1/24	N/A
	Loopback4	203.0.115.1/24	N/A
	G1/0/11	209.165.201.14/30	N/A
	G1/0/1	209.165.201.10/30	N/A

Objectives

Part 1: Build the Topology and Configure Basic Device Settings and IP Addressing

Part 2: Configure Basic eBGP and iBGP

Part 3: Configure a Peer Group for BGP MD5 Authentication

Part 4: Configure an AS-Path Filter

Part 5: Configure the Local Preference Attribute and an AS-Path Filter

Part 6: Configure the MED Attribute

Part 7: Configure ISP1 to Advertise a Default Route in BGP

Part 8: Configure ISP3 to Advertise a Summary (Aggregate) of its Prefixes

Background / Scenario

In this lab, OSPFv3 has already been configured on GW1 and GW2 to provide internal IPv4 reachability within AS 65000, including reachability between the loopback interfaces used for iBGP peering. You will build on this foundation by configuring **traditional BGP for IPv4 on all routers**, establishing iBGP and eBGP relationships and applying advanced BGP attributes and policies to control both outbound and inbound traffic behavior.

BGP may select either ISP1 or ISP2 based on default tie-breaking behavior. This can cause the installed next hop in the routing table to differ between lab runs. Where applicable this has been noted throughout the lab.

Note: Although the BGP standards (e.g., RFC 4271) define the abbreviations as **EBGP** and **IBGP**, modern vendor documentation often uses the lower-case forms **eBGP** and **iBGP** to visually distinguish “external” or

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"internal" from the core BGP protocol name. We will use lower case **e** and **i** to make is easier to visually distinguish these two protocols.

Note: The routers used with this CCNP hands-on lab is a Cisco 8200L and the two Layer 3 switches are Catalyst 9300L switches using Cisco IOS XE Release 17.12.05 (universalk9 image). Other routers and Layer 3 switches and Cisco IOS versions can be used. Depending on the model and Cisco IOS version, the commands available and the output produced might vary from what is shown in the labs.

Note: Ensure that the routers and switches have been erased and have no startup configurations. If you are unsure contact your instructor.

Required Resources

- 3 Routers (Cisco Catalyst 8200L with Cisco IOS XE Release 17.12.05c universal image or comparable)
- 2 Switches (Cisco Catalyst 9300L with Cisco IOS XE Release 17.12.05 universal image or comparable)
- Console cables to configure the Cisco IOS devices via the console ports
- Ethernet cables as shown in the topology

Instructions

Part 1: Build the Network and Configure Basic Device Settings and Interface Addressing

In Part 1, you will set up the network topology and configure basic settings and interface addressing on routers and switches.

Step 1: Cable the network as shown in the topology.

Attach the devices as shown in the topology diagram, and cable as necessary.

Step 2: Configure basic settings for each router.

a. Console into each device, enter global configuration mode, and apply the basic settings and interface addressing using the following startup configurations for each device.

Router GW1

```
hostname GW1
no ip domain lookup
line con 0
exec-timeout 0 0
logging sync
ipv6 unicast-routing
interface Loopback0
 ip address 1.1.1.1 255.255.255.255
 ipv6 enable
 ospfv3 1 ipv4 area 0
 exit
interface Loopback1
 ip address 192.168.11.1 255.255.255.0
 exit
interface GigabitEthernet0/0/0
```

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```
 description To GW2 (R2)
 ip address 172.16.12.1 255.255.255.252
 ipv6 enable
 ospfv3 1 ipv4 area 0
 no shutdown
 exit
interface GigabitEthernet0/0/1
 description To ISP1 (D1)
 ip address 209.165.201.1 255.255.255.252
 no shutdown
 exit
interface Serial0/1/0
 description To ISP2 (R3) - slower link
 ip address 209.165.201.17 255.255.255.252
 no shutdown
 exit
router ospfv3 1
 router-id 1.1.1.1
 address-family ipv4 unicast
  passive-interface default
  no passive-interface GigabitEthernet0/0/0
 exit-address-family
end
```

Router GW2

```
hostname GW2
ipv6 unicast-routing
line con 0
exec-timeout 0 0
logging sync
no ip domain lookup
interface Loopback0
 ip address 2.2.2.2 255.255.255.255
 ipv6 enable
 ospfv3 1 ipv4 area 0
 exit
interface Loopback1
 ip address 192.168.22.1 255.255.255.0
 exit
interface GigabitEthernet0/0/0
 description To GW1 (R1)
 ip address 172.16.12.2 255.255.255.252
 ipv6 enable
 ospfv3 1 ipv4 area 0
 no shutdown
 exit
```

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```
interface GigabitEthernet0/0/1
 description To ISP2 (R3) - faster link
 ip address 209.165.201.5 255.255.255.252
 no shutdown
 exit
router ospfv3 1
 router-id 2.2.2.2
 address-family ipv4 unicast
  passive-interface default
  no passive-interface GigabitEthernet0/0/0
 exit-address-family
end
```

Router ISP2

```
hostname ISP2
no ip domain lookup
line con 0
exec-timeout 0 0
logging sync
interface GigabitEthernet0/0/0
 description To GW2 (R2)
 ip address 209.165.201.6 255.255.255.252
 no shutdown
 exit
interface Serial0/1/0
 description To GW1 (R1) - slower link
 ip address 209.165.201.18 255.255.255.252
 no shutdown
 exit
interface GigabitEthernet0/0/1
 description To ISP3 (D2)
 ip address 209.165.201.13 255.255.255.252
 no shutdown
 exit
end
```

Switch ISP1

```
hostname ISP1
no ip domain lookup
line con 0
exec-timeout 0 0
logging sync
ip routing
interface Loopback0
 ip address 209.165.201.21 255.255.255.255
 exit
```

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```
interface GigabitEthernet1/0/11
 no switchport
 description To GW1 (R1)
 ip address 209.165.201.2 255.255.255.252
 no shutdown
 exit
interface GigabitEthernet1/0/1
 no switchport
 description To ISP3 (D2)
 ip address 209.165.201.9 255.255.255.252
 no shutdown
 exit
end
```

Switch ISP3

```
hostname ISP3
no ip domain lookup
line con 0
exec-timeout 0 0
logging sync
ip routing
interface Loopback1
 ip address 203.0.112.1 255.255.255.0
 exit
interface Loopback2
 ip address 203.0.113.1 255.255.255.0
 exit
interface Loopback3
 ip address 203.0.114.1 255.255.255.0
 exit
interface Loopback4
 ip address 203.0.115.1 255.255.255.0
 exit
interface GigabitEthernet1/0/11
 no switchport
 description To ISP2 (R3)
 ip address 209.165.201.14 255.255.255.252
 no shutdown
 exit
interface GigabitEthernet1/0/1
 no switchport
 description To ISP1 (D1)
 ip address 209.165.201.10 255.255.255.252
 no shutdown
 exit
end
```

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b. Save the running configuration to startup-config.

Part 2: Configure Basic eBGP and iBGP

In this part, you will review the existing OSPFv3 configuration on the gateway routers to confirm internal IPv4 reachability and verify that loopback interfaces used for iBGP peering are correctly learned through OSPFv3. You will then configure a combination of iBGP and eBGP sessions across the topology, using loopback-based iBGP between gateway routers for resiliency and eBGP connections to multiple ISPs to simulate external connectivity.

You will advertise local prefixes into BGP, verify that routes are exchanged successfully across all autonomous systems, and confirm end-to-end reachability by examining IPv4 routing tables. Finally, you will explore how 16-bit and 32-bit AS numbers coexist in BGP, including how Cisco devices display and convert between **asplain** and **asdot** notation, reinforcing both practical configuration skills and conceptual understanding of modern BGP operation.

Step 1: Examine commands used for OSPFv3 in GW1 and GW2 running-configs.

Before configuring BGP, examine the existing OSPFv3 configuration on **GW1** and **GW2**. OSPFv3 has already been configured to provide internal reachability between the gateway routers. This step is intended to familiarize you with how OSPFv3 is enabled and verified in the running configuration before proceeding with the BGP configuration.

```
GW1# show running-config brief
Building configuration...
<output omitted>
!
ipv6 unicast-routing
!
<output omitted>
!
interface Loopback0
 ip address 1.1.1.1 255.255.255.255
 ipv6 enable
 ospfv3 1 ipv4 area 0
!
<output omitted>
!
interface GigabitEthernet0/0/0
 description To GW2 (R2)
 ip address 172.16.12.1 255.255.255.252
 negotiation auto
 ipv6 enable
 ospfv3 1 ipv4 area 0
!
<output omitted>
!
router ospfv3 1
 router-id 1.1.1.1
 !
 address-family ipv4 unicast
  passive-interface default
  no passive-interface GigabitEthernet0/0/0
```

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```
 exit-address-family
!
<output omitted>
end

GW2# show running-config brief
Building configuration...
!
<output omitted>
!
ipv6 unicast-routing
!
<output omitted>
!
interface Loopback0
 ip address 2.2.2.2 255.255.255.255
 ipv6 enable
 ospfv3 1 ipv4 area 0
!
<output omitted>
!
interface GigabitEthernet0/0/0
 description To GW1 (R1)
 ip address 172.16.12.2 255.255.255.252
 negotiation auto
 ipv6 enable
 ospfv3 1 ipv4 area 0
!
<output omitted>
!
router ospfv3 1
 router-id 2.2.2.2
 !
 address-family ipv4 unicast
  passive-interface default
  no passive-interface GigabitEthernet0/0/0
 exit-address-family
!
<output omitted>
end
```

Step 2: Verify the IPv4 routing tables for OSPFv3 routes.

Examine the IPv4 routing table on each router and confirm that the **Loopback0 address of the other gateway** has been learned via OSPFv3. The presence of an OSPF-learned /32 route confirms that OSPFv3 is correctly exchanging IPv4 routing information and that the loopback interfaces used for iBGP peering are reachable.

At this point, no default route should be present, and only the internal OSPFv3 routes should appear in the routing table.

```
GW1# show ip route ospfv3 | begin Gateway
```

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```
Gateway of last resort is not set

      2.0.0.0/32 is subnetted, 1 subnets
O        2.2.2.2 [110/1] via 172.16.12.2, 00:13:33, GigabitEthernet0/0/0

GW2# show ip route ospfv3 | begin Gateway
Gateway of last resort is not set

      1.0.0.0/32 is subnetted, 1 subnets
O        1.1.1.1 [110/1] via 172.16.12.1, 00:14:42, GigabitEthernet0/0/0
```

Step 3: Configure BGP (iBGP-eBGP) on GW1.

In this step, you will configure **BGP on GW1** to establish both **internal BGP (iBGP)** and **external BGP (eBGP)** neighbor relationships.

The iBGP session between **GW1 and GW2** is configured using the **Loopback0 interfaces** with the **update-source loopback0** command. Using loopback addresses for iBGP provides a more resilient and realistic design, as loopbacks remain reachable even if the physical path between routers changes. In a real-world topology, there would typically be **multiple paths between gateway routers**, and using loopbacks ensures the iBGP session is not tied to a single physical interface.

Because GW1 and GW2 are acting as gateway routers and exchange routes learned from external eBGP peers via iBGP, the **next-hop-self** command must be configured on both routers. Without **next-hop-self**, routes learned from an eBGP neighbor would retain their original external next-hop address when advertised via iBGP

GW1 also establishes eBGP sessions with **ISP1** and **ISP2**, representing connections to external autonomous systems. Finally, GW1 advertises the **192.168.11.0/24** network into BGP, making this local prefix reachable by external peers.

```
GW1(config)# router bgp 65000
GW1(config-router)# bgp log-neighbor-changes
GW1(config-router)# ! iBGP to GW2 using loopbacks
GW1(config-router)# neighbor 2.2.2.2 remote-as 65000
GW1(config-router)# neighbor 2.2.2.2 update-source Loopback0
GW1(config-router)# neighbor 2.2.2.2 next-hop-self
GW1(config-router)# ! eBGP to ISP1 (directly connected)
GW1(config-router)# neighbor 209.165.201.2 remote-as 65100
GW1(config-router)# ! eBGP to ISP2 over the serial link
GW1(config-router)# neighbor 209.165.201.18 remote-as 65200
GW1(config-router)# ! advertise local prefixes (from AS 65000)
GW1(config-router)# network 192.168.11.0 mask 255.255.255.0
GW1(config-router)# end
```

Step 4: Configure BGP (iBGP-eBGP) on GW2.

This step mirrors the BGP configuration performed on **GW1**. GW2 is configured with an **iBGP session to GW1** using the **Loopback0 interface** for resiliency, along with an **eBGP session to ISP2**. GW2 then advertises its local prefix, **192.168.22.0/24**, into BGP.

```
GW2(config)# router bgp 65000
GW2(config-router)# bgp log-neighbor-changes
GW2(config-router)# ! iBGP to GW1 using loopbacks
```

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```
GW2(config-router)# neighbor 1.1.1.1 remote-as 65000
GW2(config-router)# neighbor 1.1.1.1 update-source Loopback0
*Jan 26 18:12:58.172: %BGP-5-ADJCHANGE: neighbor 1.1.1.1 Up
GW2(config-router)# neighbor 1.1.1.1 next-hop-self
GW2(config-router)# ! eBGP to ISP2 (directly connected)
GW2(config-router)# neighbor 209.165.201.6 remote-as 65200
GW2(config-router)# ! advertise local prefixes (from AS 65000)
GW2(config-router)# network 192.168.22.0 mask 255.255.255.0
GW2(config-router)# end
```

Step 5: Configure BGP (eBGP) on ISP2.

In this step, **ISP2** is configured with **external BGP (eBGP)** sessions to both gateway routers in **AS 65000** as well as to **ISP3**. These peerings allow ISP2 to participate in external route exchange and later serve as a policy decision point for traffic entering and leaving AS 65000.

```
ISP2(config)# router bgp 65200
ISP2(config-router)# bgp log-neighbor-changes
ISP2(config-router)# ! BGP router-id so ISP2 is preferred over ISP1
ISP2(config-router)# bgp router-id 4.4.4.4
ISP2(config-router)# ! eBGP to GW2
ISP2(config-router)# neighbor 209.165.201.5 remote-as 65000
ISP2(config-router)# ! eBGP to GW1 (serial)
*Jan 26 18:26:20.078: %BGP-5-ADJCHANGE: neighbor 209.165.201.5 Up
*Jan 26 18:26:20.323: %BGP-5-ADJCHANGE: neighbor 209.165.201.17 Up
ISP2(config-router)# neighbor 209.165.201.17 remote-as 65000
ISP2(config-router)# ! eBGP to ISP3
ISP2(config-router)# neighbor 209.165.201.14 remote-as 65300.3
ISP2(config-router)# end
```

Step 6: Configure BGP (eBGP) on ISP3.

In this step, **ISP3** is configured as an external autonomous system using a **32-bit ASN**. ISP3 establishes **eBGP sessions with ISP1 and ISP2** and originates four IPv4 prefixes (**203.0.112.0/24 through 203.0.115.0/24**) into BGP. These routes represent external networks and will later be used to demonstrate BGP policy behavior and route aggregation.

Note: 32-bit ASNs will be explained in a later step.

```
ISP3(config)# router bgp 65300.3
ISP3(config-router)# bgp log-neighbor-changes
ISP3(config-router)# ! eBGP to ISP1
ISP3(config-router)# neighbor 209.165.201.9 remote-as 65100
ISP3(config-router)# ! eBGP to ISP2
ISP3(config-router)# neighbor 209.165.201.13 remote-as 65200
*Jan 26 13:17:56.176: %BGP-5-ADJCHANGE: neighbor 209.165.201.13 Up
ISP3(config-router)# ! originate the /24s (aggregation comes later)
ISP3(config-router)# network 203.0.112.0 mask 255.255.255.0
ISP3(config-router)# network 203.0.113.0 mask 255.255.255.0
ISP3(config-router)# network 203.0.114.0 mask 255.255.255.0
ISP3(config-router)# network 203.0.115.0 mask 255.255.255.0
```

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```
ISP3(config-router)# end
```

Step 7: Configure eBGP on ISP1.

In this step, **ISP1** is configured with **external BGP (eBGP)** sessions to both the **customer AS (65000)** and **ISP3**. These eBGP peerings allow ISP1 to exchange routing information between the customer network and the upstream provider. This step must be performed only after GW1 has learned the ISP3 prefixes via ISP2, ensuring that ISP2 is the initially selected and oldest path using **show ip route bgp**.

```
GW1# show ip route bgp | begin Gateway
Gateway of last resort is not set

B     192.168.22.0/24 [200/0] via 2.2.2.2, 00:22:06
B     203.0.112.0/24 [20/0] via 209.165.201.18, 00:20:40
B     203.0.113.0/24 [20/0] via 209.165.201.18, 00:20:40
B     203.0.114.0/24 [20/0] via 209.165.201.18, 00:20:40
B     203.0.115.0/24 [20/0] via 209.165.201.18, 00:20:40
GW1#
```

After GW1 has ISP3 prefixes showing 2.2.2.2 (GW2) as the next-hop router, configure BGP on ISP1.

```
ISP1(config)# router bgp 65100
ISP1(config-router)# bgp log-neighbor-changes
ISP1(config-router)# ! BGP router-id so ISP2 is preferred over ISP1
ISP1(config-router)# bgp router-id 9.9.9.9
ISP1(config-router)# ! eBGP to customer AS (65000)
ISP1(config-router)# neighbor 209.165.201.1 remote-as 65000
ISP1(config-router)# ! eBGP to ISP3 (65300.3)
ISP1(config-router)# neighbor 209.165.201.10 remote-as 65300.3
*Jan 26 13:15:22.067: %BGP-5-ADJCHANGE: neighbor 209.165.201.1 Up
ISP1(config-router)# end
```

Note on Configuration Sequence: In this lab, BGP is intentionally configured on all routers except ISP1 first to ensure deterministic behavior. Before ISP1 participates in BGP, GW1 learns the ISP3 prefixes only via ISP2, making that path the oldest and initially installed route. When ISP1 is later configured and begins advertising the same prefixes, all higher-priority BGP attributes (weight, local preference, AS path length, origin, and MED) are equal, so the BGP decision process falls to its final tie-breaking steps. Because the path via ISP2 was learned first and ISP2 has a lower configured BGP router ID than ISP1, GW1 continues to prefer ISP2 as the outbound path.

Note on Router-IDs: BGP router IDs are explicitly configured on ISP1 and ISP2 to provide another deterministic tie-breaker in the BGP best-path selection process, after oldest path The configured BGP router IDs on ISP1 and ISP2 are not the primary reason for path selection at this point. Instead, they serve as a fallback mechanism that would be used only if the paths were received at the same time and remained otherwise indistinguishable (oldest path not a factor). In that case, BGP would select the path from the peer with the lower router ID.

Important: In real-world networks, autonomous systems do not typically rely on oldest-path or a peer's router ID to influence routing decisions. Oldest-path or router ID are used only as a last-resort tie-breaker as they provide no policy control. Production networks instead use attributes such as Local Preference (for outbound traffic control) and MED (for influencing inbound traffic) to enforce intentional routing policy. These attributes will be configured and explored later in this lab.

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Step 8: Examine IPv4 routing tables and verify reachability.

In this step, examine the **IPv4 BGP entries in the routing tables** on each router to verify that **all autonomous systems have reachability to every originated BGP prefix**. The presence of these routes confirms that iBGP and eBGP sessions are established correctly and that BGP is successfully exchanging routing information across the topology. Before examining the routing tables, note which prefixes are originated by each AS.

AS 65000 (GW1 and GW2)

- GW1 originates: 192.168.11.0/24
- GW2 originates: 192.168.22.0/24

AS 65300.3 (ISP3 - 32-bit ASN) originates

- 203.0.112.0/24
- 203.0.113.0/24
- 203.0.114.0/24
- 203.0.115.0/24

AS 65100 (ISP1)

- Does not originate customer prefixes in this step

AS 65200 (ISP2)

- Does not originate customer prefixes in this step.

At this point, all BGP-originated prefixes, whether learned via eBGP or iBGP, should be present in the IPv4 routing tables of every router in the topology, confirming full end-to-end reachability.

Note: At this stage of the lab, no outbound BGP policy has been configured on GW1. As a result, multiple external paths to the same destination may appear equivalent, and **BGP may select either ISP1 or ISP2** based on default tie-breaking behavior. This can cause the installed next hop in the routing table to differ between lab runs. This behavior is expected and demonstrates the need for **Local Preference** (later in this lab) to enforce a consistent outbound path selection policy.

```
GW1# show ip route bgp | begin Gateway
Gateway of last resort is not set

B     192.168.22.0/24 [200/0] via 2.2.2.2, 00:05:28
B     203.0.112.0/24 [20/0] via 209.165.201.18, 00:05:15
B     203.0.113.0/24 [20/0] via 209.165.201.18, 00:05:15
B     203.0.114.0/24 [20/0] via 209.165.201.18, 00:05:15
B     203.0.115.0/24 [20/0] via 209.165.201.18, 00:05:15

GW2# show ip route bgp | begin Gateway
Gateway of last resort is not set

B     192.168.11.0/24 [200/0] via 1.1.1.1, 00:05:17
B     203.0.112.0/24 [20/0] via 209.165.201.6, 00:05:04
B     203.0.113.0/24 [20/0] via 209.165.201.6, 00:05:04
B     203.0.114.0/24 [20/0] via 209.165.201.6, 00:05:04
B     203.0.115.0/24 [20/0] via 209.165.201.6, 00:05:04

ISP1# show ip route bgp | begin Gateway
Gateway of last resort is not set
```

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```
B     192.168.11.0/24 [20/0] via 209.165.201.1, 00:12:58
B     192.168.22.0/24 [20/0] via 209.165.201.1, 00:12:40
B     203.0.112.0/24 [20/0] via 209.165.201.10, 00:12:58
B     203.0.113.0/24 [20/0] via 209.165.201.10, 00:12:58
B     203.0.114.0/24 [20/0] via 209.165.201.10, 00:12:58
B     203.0.115.0/24 [20/0] via 209.165.201.10, 00:12:58

ISP2# show ip route bgp | begin Gateway
Gateway of last resort is not set

B     192.168.11.0/24 [20/0] via 209.165.201.17, 00:13:24
B     192.168.22.0/24 [20/0] via 209.165.201.17, 00:13:24
B     203.0.112.0/24 [20/0] via 209.165.201.14, 00:13:24
B     203.0.113.0/24 [20/0] via 209.165.201.14, 00:13:24
B     203.0.114.0/24 [20/0] via 209.165.201.14, 00:13:24
B     203.0.115.0/24 [20/0] via 209.165.201.14, 00:13:24

ISP3# show ip route bgp | begin Gateway bgp
Gateway of last resort is not set

B     192.168.11.0/24 [20/0] via 209.165.201.13, 00:13:49
B     192.168.22.0/24 [20/0] via 209.165.201.13, 00:13:48
```

Step 9: Understanding 32-bit ASNs.

a. Originally, BGP AS numbers were **16-bit values**, which limited the Internet to about **65,000 ASes**. As the Internet grew and more organizations became multihomed, this space was exhausted. To solve this, **32-bit AS numbers** were introduced, expanding the AS number space dramatically while remaining compatible with older routers.

Modern **network operating systems** fully support **both 16-bit and 32-bit ASNs**, allowing them to coexist within the same BGP AS path without affecting routing behavior.

Representing 32-bit ASNs

Although a 32-bit AS number is assigned as a single numeric value, it can be displayed in two different formats. Cisco devices support both formats, and the choice affects only how the AS number is shown, not how BGP operates.

- **asplain format**: Displays the AS number as a single 32-bit integer (for example, **4279500803**).
- **asdot format**: Displays the same 32-bit AS number as two 16-bit values separated by a dot (for example, **65300.3**).

Both formats represent the same AS number, and routing behavior is identical regardless of which format is used.

The following BGP output shows how **16-bit and 32-bit ASNs coexist in the same AS path**. In this example, **65200** represents a traditional 16-bit ASN, while **4279500803** represents a 32-bit ASN in **asplain** format. Modern BGP implementations handle both seamlessly, and the presence of both values in the AS path does not affect route selection or reachability.

```
GW1# show ip bgp
BGP table version is 7, local router ID is 192.168.11.1
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
```

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```
              r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,
              x best-external, a additional-path, c RIB-compressed,
              t secondary path, L long-lived-stale,
Origin codes: i - IGP, e - EGP, ? - incomplete
RPKI validation codes: V valid, I invalid, N Not found

     Network          Next Hop            Metric LocPrf Weight Path
*>   192.168.11.0     0.0.0.0                  0         32768 i
*>i  192.168.22.0     2.2.2.2                  0    100      0 i
*    203.0.112.0      209.165.201.2                         0 65100 4279500803 i
* i                   2.2.2.2                  0    100      0 65200 4279500803 i
*>                   209.165.201.18                        0 65200 4279500803 i
<output omitted>

GW1# show ip bgp 203.0.112.0 bestpath
BGP routing table entry for 203.0.112.0/24, version 8
Paths: (3 available, best #3, table default)
  Advertised to update-groups:
     1          2
  Refresh Epoch 1
  65200 4279500803
    209.165.201.18 from 209.165.201.18 (4.4.4.4)
      Origin IGP, localpref 100, valid, external, best
      rx pathid: 0, tx pathid: 0x0
      Updated on Jan 29 2026 12:36:11 UTC
```

b. By default, Cisco routers display 32-bit AS numbers in **asplain format**, which shows the AS as a single large integer. The **bgp asnotation dot** command changes **only how 32-bit ASNs are displayed**, not how BGP operates.

   After enabling dot notation, the same 32-bit ASN that previously appeared as **4279500803** is now shown in **asdot format** as **65300.3**. This format splits the 32-bit value into two 16-bit components, making the AS path easier to read and interpret

```
GW1(config)# router bgp 65000
GW1(config-router)# bgp asnotation dot
GW1(config-router)# end
GW1# show ip bgp
BGP table version is 11, local router ID is 192.168.11.1
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
              r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,
              x best-external, a additional-path, c RIB-compressed,
              t secondary path, L long-lived-stale,
Origin codes: i - IGP, e - EGP, ? - incomplete
RPKI validation codes: V valid, I invalid, N Not found

     Network          Next Hop            Metric LocPrf Weight Path
*>   192.168.11.0     0.0.0.0                  0         32768 i
*>i  192.168.22.0     2.2.2.2                  0    100      0 i
*    203.0.112.0      209.165.201.2                         0 65100 65300.3 i
* i                   2.2.2.2                  0    100      0 65200 65300.3 i
```

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```
*>                    209.165.201.18                         0 65200 65300.3 i
<output omitted>
```

Note: Use the **no bgp asnotation dot** command to return to the default **asplain** format.

c. **Converting from asplain to asdot Notation**.

A 32-bit AS number (**asplain**) is a single numeric value. To display it in **asdot** notation, the number is split into two 16-bit components.

To convert **asplain** (**4279500803**) to **asdot**:

1) Divide the 32-bit AS number by 65,536

   4279500803 / 65,536 = **65,300** remainder **3**

2) The quotient becomes the first value (X)

   X = 65000

3) The remainder becomes the second value (Y)

   Y = 3

Resulting in the **asdot** notation: **65000.3**

d. **Converting from asdot to asplain Notation**.

An AS number written in **asdot** format represents two 16-bit values joined together. To convert it back to the **asplain** (single 32-bit) value, the two components are recombined mathematically.

To convert **asdot** (**65300.3**) to **asplain**:

1) Multiply the first value (X) by 65,536

   65,300 x 65,536 = 4,279,500,800

2) Add the second value (Y)

   4,279,500,800 **+ 3** = **4,279,500,803**

Resulting in the **asplain** notation: **4279500803**

e. **Private 32-bit AS Number Ranges**.

In addition to the traditional 16-bit private AS range, a large block of **32-bit AS numbers** is also reserved for **private use** and is not advertised on the public Internet.

asplain format:

- 4200000000 – 4294967294

asdot format (equivalent):

- 64086.64086 – 65535.65534

These private 32-bit ASNs function exactly like public ASNs in BGP but are intended for **lab environments, private networks, and internal routing designs**. As with private 16-bit ASNs, they should not be used on the public Internet.

Part 3: Configure a Peer Group for BGP MD5 Authentication

In networks where a router peers with **multiple BGP neighbors** that share common settings (timers, authentication, inbound/outbound policy, etc.), configuring each neighbor individually can become repetitive and error-prone. A **BGP peer group** allows you to define these shared settings **once** and then apply them to multiple neighbors. Each neighbor still keeps its own unique properties (such as **remote-as** and neighbor IP), but inherits the common settings from the peer group.

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In this lab, **GW1** is multihomed to two different ISPs (**ISP1 and ISP2**), making it an ideal place to use a peer group. You will create a peer group on GW1 and apply it to both eBGP neighbors. In the next part, you will use **BGP MD5 authentication** as a real-world example of why peer groups are useful.

Step 1: Configure a peer group on GW1.

Create a peer group on GW1 for its eBGP ISP neighbors and apply the peer group to:

- ISP1 neighbor 209.165.201.2 (AS 65100)
- ISP2 neighbor 209.165.201.18 (AS 65200)

Note: A peer group can include neighbors in different ASes. The **remote-as** is still configured per neighbor, while common settings are applied through the peer group.

```
GW1(config)# router bgp 65000
GW1(config-router)# ! Create the peer group
GW1(config-router)#  neighbor ISP-PEERS peer-group
GW1(config-router)# ! Identify each eBGP neighbor and its remote AS
GW1(config-router)#  neighbor 209.165.201.2 remote-as 65100
GW1(config-router)#  neighbor 209.165.201.18 remote-as 65200
GW1(config-router)# ! Apply the peer group to each neighbor
GW1(config-router)#  neighbor 209.165.201.2 peer-group ISP-PEERS
GW1(config-router)#  neighbor 209.165.201.18 peer-group ISP-PEERS
GW1(config-router)# end
```

Step 2: Verify that both neighbors are present and established.

Confirm that both neighbors are still present and established. In the **show ip bgp summary** output, a numeric value in the **State/PfxRcd** column indicates that the BGP session is established and shows the number of prefixes received from that neighbor.

```
GW1# show ip bgp summary
BGP router identifier 192.168.11.1, local AS number 65000
<output omitted>

Neighbor        V      AS MsgRcvd MsgSent   TblVer  InQ OutQ Up/Down  State/PfxRcd
2.2.2.2         4   65000      32      32        7    0    0 00:25:21        5
209.165.201.2   4   65100      32      33        7    0    0 00:25:09        4
209.165.201.18  4   65200      33      32        7    0    0 00:25:00        4
```

Step 3: Configure BGP MD5 authentication.

BGP sessions run over TCP, and by default they are **not authenticated**. BGP supports **MD5 authentication**, which adds a shared secret to the TCP session to protect against unauthorized peers and spoofed connections.

When MD5 authentication is configured on a **BGP peer group**, the authentication settings are applied consistently to all neighbors that belong to that peer group. This is a common real-world use case for peer groups and mirrors the configuration approach used in the Cisco reference lab.

In this step, you will configure **MD5 authentication** between **GW1** and its external BGP neighbors **ISP1** and **ISP2**.

On **GW1**, apply the MD5 password to the existing peer group. This **single command** enables authentication for **both eBGP neighbors** in the group. A BGP session may not visibly reset when MD5 authentication is

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added if the matching password is configured quickly on both peers; authentication is verified by the session remaining established.

```
GW1(config)# router bgp 65000
GW1(config-router)# ! Apply MD5 authentication to the peer group
GW1(config-router)# neighbor ISP-PEERS password cisco123
GW1(config-router)# end
```

Step 4: Configure matching MD5 authentication on ISP1.

ISP1 must be configured with the **same password** for the BGP session with GW1.

```
ISP1(config)# router bgp 65100
ISP1(config-router)# ! Match MD5 password for GW1
ISP1(config-router)# neighbor 209.165.201.1 password cisco123
ISP1(config-router)# end
```

Step 5: Configure matching MD5 authentication on ISP2.

ISP2 must also be configured with the **same password** for its BGP session with GW1.

```
ISP2(config)# router bgp 65200
ISP2(config-router)# ! Match MD5 password for GW1
ISP2(config-router)#  neighbor 209.165.201.17 password cisco123
ISP2(config-router)#  end
```

Step 6: Verifying peer group membership and BGP session state.

The output confirms that both eBGP neighbors on **GW1** are **members of the peer group ISP-PEERS**, indicating that they inherit the shared peer-group settings, including MD5 authentication. The **BGP state = Established** for each neighbor verifies that the authenticated BGP sessions are up and stable and that GW1 is successfully exchanging routing information with both ISP neighbors.

```
GW1# show ip bgp neighbor 209.165.201.2
BGP neighbor is 209.165.201.2,  remote AS 65100, external link
 Member of peer-group ISP-PEERS for session parameters
  BGP version 4, remote router ID 9.9.9.9
  BGP state = Established, up for 00:28:13
  Last read 00:00:22, last write 00:00:47, hold time is 180, keepalive interval is 60
seconds
  Last update received: 00:26:32
<output omitted>
GW1# show ip bgp neighbor 209.165.201.18
BGP neighbor is 209.165.201.18,  remote AS 65200, external link
 Member of peer-group ISP-PEERS for session parameters
  BGP version 4, remote router ID 4.4.4.4
  BGP state = Established, up for 00:29:02
  Last read 00:00:48, last write 00:00:56, hold time is 180, keepalive interval is 60
seconds
  Last update received: 00:27:58
<output omitted>
```

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Part 4: Configure an AS-Path Filter (Prevent AS 65000 from Becoming a Transit AS)

At this point in the lab, **GW1 and GW2 are fully multihomed**, with BGP connectivity to multiple service providers. While this provides redundancy, it also introduces an **unintended side effect**: without additional policy controls, **ISP1 and ISP2 could potentially use AS 65000 as a transit path** to reach other external networks if a direct path fails.

A **transit AS** is an autonomous system that forwards traffic between other autonomous systems even though it is neither the source nor the destination of that traffic.

In real-world deployments, **customer networks do not act as transit autonomous systems** for their providers. AS 65000 should advertise **only its own locally originated prefixes** and should never forward traffic between ISPs.

To enforce this behavior, AS 65000 will use an **outbound AS-path filter**. Locally originated routes are easily identified because they have an **empty AS-path attribute**. By permitting only routes with an empty AS path and denying all others, AS 65000 ensures that it advertises **only its own prefixes** and does not become a transit AS.

AS-path filters use **AS-path access lists**, which rely on **regular expressions** to match patterns in the AS-path attribute. In this lab, you will use a simple and commonly deployed expression to allow only locally sourced routes.

Commonly Used AS-Path Regular Expression Characters

Symbol	Meaning
^	Beginning of the AS path
$	End of the AS path
_	AS delimiter (beginning, end, or space between AS numbers)
.	Any single character
.*	Any number of characters (match anything)
+	One or more occurrences of the previous pattern
*****	Zero or more occurrences of the previous pattern
?	Zero or one occurrence of the previous pattern
****	Escapes a special character
[ ]	Matches a range or set of characters

Note: In BGP AS-path filtering, AS numbers are treated as a **string**, not as numeric values.

Common AS-Path Regular Expression Examples

Regular Expression	Description
^$	Matches **locally originated routes only** (empty AS path)

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Regular Expression	Description
^65000$	Matches routes originating in **AS 65000**
_65000$	Matches routes **originating** in AS 65000
^65000_	Matches routes **received from** AS 65000
_65000_	Matches routes that **passed through** AS 65000
.*	Matches **all routes**
[0-9]+	Matches any AS number
([0-9]+)(_\1)*	Matches an AS number repeated any number of times (AS-path prepending)
^65100_65300\.3$	Matches a **specific AS path sequence**

Step 1: Verify AS 65000 is currently a transit AS.

The BGP table on ISP1 shows that AS 65000 is currently functioning as a transit AS, meaning it is advertising routes learned from one external AS to another external AS.

The BGP table on ISP1 shows possible routes where the **first AS in the AS path is 65000**, followed by other AS numbers. This indicates that the routes were **learned from AS 65000**, even though they **originated in a different AS**. Because AS 65000 is advertising routes that it learned from one external provider to another external provider, it is functioning as a **transit AS**, allowing traffic to pass through its network rather than advertising only its own locally originated prefixes.

Note: Earlier in this lab we stated that no outbound BGP policy has been configured on GW1. As a result, multiple external paths to the same destination may appear equivalent, and **BGP may select either ISP1 or ISP2** based on default tie-breaking behavior. Because of this, AS 65000 may appear as a transit AS when examining the BGP table on either ISP1 or ISP2, depending on which path is currently preferred. **If AS 65000 does not appear as the first AS followed by other AS numbers** in the AS path on ISP1, you should **examine the BGP table on ISP2 instead**.

```
ISP1# show ip bgp
BGP table version is 7, local router ID is 209.165.201.21
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
              r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,
              x best-external, a additional-path, c RIB-compressed,
              t secondary path, L long-lived-stale,
Origin codes: i - IGP, e - EGP, ? - incomplete
RPKI validation codes: V valid, I invalid, N Not found

     Network          Next Hop            Metric LocPrf Weight Path
 *>   192.168.11.0     209.165.201.1            0             0 65000 i
 *    192.168.22.0     209.165.201.10                         0 4279500803 65200 65000 i
 *>                    209.165.201.1                          0 65000 i
 *    203.0.112.0      209.165.201.1                          0 65000 65200 4279500803 i
 *>                    209.165.201.10           0             0 4279500803 i
 *    203.0.113.0      209.165.201.1                          0 65000 65200 4279500803 i
 *>                    209.165.201.10           0             0 4279500803 i
```

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```
 *    203.0.114.0      209.165.201.1                          0 65000 65200 4279500803 i
 *>                    209.165.201.10         0               0 4279500803 i
*    203.0.115.0      209.165.201.1                          0 65000 65200 4279500803 i
 *>                    209.165.201.10         0               0 4279500803 i
```

Step 2: Configure an outbound AS-Path filter on GW2 (prevent transit).

GW2 will be configured to advertise only locally originated routes to its ISP **neighbor (ISP2)**. This prevents AS 65000 from acting as a transit AS. Applying the filter-list with **out** controls what GW2 **advertises to ISP2**, allowing only GW2's locally originated prefixes and preventing it from advertising routes learned from other neighbors (such as via iBGP from GW1).

AS-path regular expression: ^$

The AS-path access list uses a regular expression to match the AS-path attribute:

- **^** = beginning of the AS-path string
- **$** = end of the AS-path string
- **^$** therefore matches an empty AS path

An empty AS path means the route was **originated locally on this router** (for example, via a network statement). Routes learned from eBGP neighbors will have one or more AS numbers in the path and **will not** match ^$.

Why apply the filter-list out

Applying the filter-list with **out** means the filter controls **what GW2 advertises to its neighbor**. In other words:

- GW2 will advertise its own prefixes
- GW2 will not advertise routes learned from **other neighbors** (such as via iBGP from GW1) **to ISP2**.
- This prevents AS 65000 from being used as a transit path

```
GW2(config)# ! Permit only locally originated routes (empty AS-path)
GW2(config)# ip as-path access-list 1 permit ^$
GW2(config)# router bgp 65000
GW2(config-router)# ! Apply outbound AS-path filter to the ISP neighbor
GW2(config-router)# neighbor 209.165.201.6 filter-list 1 out
GW2(config-router)# end
```

Step 3: Configure an outbound AS-Path filter on GW1 (prevent transit).

In this step, GW1 is configured to advertise **only locally originated routes** to its external BGP neighbors. Because **GW1 uses a peer group for its ISP neighbors**, the AS-path filter can be applied **once to the peer group** and automatically inherited by **both ISP1 and ISP2**.

```
GW1(config)# ! Permit only locally originated routes (empty AS-path)
GW1(config)# ip as-path access-list 1 permit ^$
GW1(config)# router bgp 65000
GW1(config-router)# ! Apply outbound AS-path filter to the ISP peer group
GW1(config-router)# neighbor ISP-PEERS filter-list 1 out
GW1(config-router)# end
```

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Step 4: Verify AS 65000 is no longer a transit AS.

After applying the outbound AS-path filters on **GW1 and GW2**, re-examine the BGP table on **ISP1** (or ISP2 if applicable) again.

Notice **AS Path** column for the external prefixes (**203.0.112.0/24–203.0.115.0/24**). These routes now show an AS path of: **4279500803**. (AS **4279500803** is shown in **asplain** notation because we did not use the BGP **bgp asnotation dot** command on ISP1.)

This indicates that the routes are learned **directly from ISP3**, without passing through **AS 65000**. The absence of **65000** in the AS path for these external routes confirms that AS 65000 is **no longer advertising routes learned from one ISP to another**.

In contrast, the customer prefixes (**192.168.11.0/24** and **192.168.22.0/24**) still include **65000** in the AS path, which is expected because these routes originate in AS 65000.

In other words, if AS 65000 does not appear in an AS path **followed by other AS numbers** for routes that originate in different ASes, then AS 65000 is no longer acting as a transit AS.

Note: It may take a moment for BGP to reconverge.

```
ISP1# show ip bgp
BGP table version is 7, local router ID is 9.9.9.9
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
              r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,
              x best-external, a additional-path, c RIB-compressed,
              t secondary path, L long-lived-stale,
Origin codes: i - IGP, e - EGP, ? - incomplete
RPKI validation codes: V valid, I invalid, N Not found

     Network          Next Hop            Metric LocPrf Weight Path
 *>   192.168.11.0     209.165.201.1            0             0 65000 i
 *    192.168.22.0     209.165.201.10                         0 4279500803 65200 65000 i
 *>                    209.165.201.1                          0 65000 i
 *>   203.0.112.0      209.165.201.10           0             0 4279500803 i
 *>   203.0.113.0      209.165.201.10           0             0 4279500803 i
 *>   203.0.114.0      209.165.201.10           0             0 4279500803 i
 *>   203.0.115.0      209.165.201.10           0             0 4279500803 i
```

Part 5: Configure the Local Preference Attribute and an AS-Path Filter

This part defines outbound traffic policy from the perspective of GW1, which acts as the primary exit point for AS 65000 and shares its routing decisions with GW2 using iBGP.

AS **65000** is a multihomed customer connected to multiple service providers. Although AS 65000 has multiple external paths, it has **clear outbound traffic preferences** based on link capacity and design intent:

- Primary exit: GW1 → ISP1 (fastest and most preferred link)
- Secondary exit: GW1 → ISP2
- Tertiary / last-resort exit: GW1 → GW2 → ISP2 (used only when both GW1 uplinks are unavailable)

To enforce this policy, AS 65000 must influence **outbound path selection**, not inbound traffic. This is accomplished by manipulating the **Local Preference** attribute, which is evaluated **inside the AS** and is not advertised to external neighbors.

In addition, AS 65000 must **remain a non-transit AS**, meaning it should not carry traffic between its ISPs. An **AS-path filter** is used alongside Local Preference to ensure only appropriate routes are selected and advertised.

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Why Local Preference Is the Correct Tool Here

- Local Preference controls **which exit point is chosen for outbound traffic**
- Higher Local Preference = **more preferred path**
- It is evaluated **before AS-path length** in the BGP decision process
- It applies **only within AS 65000**, making it ideal for customer policy

This aligns perfectly with our design goal: “Prefer ISP1 first, then ISP2 via GW1, and only use GW2 → ISP2 if everything else fails.”

How This Maps to Our Topology

Path	Role	Local Preference
GW1 → ISP1 (AS 65100)	Primary	**200**
GW1 → ISP2 (AS 65200)	Secondary	**150**
GW1 → GW2 → ISP2 (AS 65200)	Backup	**100 (default)**

Because GW1 and GW2 exchange routes via **iBGP**, Local Preference set on GW1 is propagated to GW2 automatically.

Step 1: Match routes learned from ISP1 (AS 65100).

On **GW1**, create an AS-path access list that matches routes **received from ISP1**.

```
GW1(config)# ip as-path access-list 10 permit ^65100_
```

Explanation:

- **^65100_** matches routes where 65100 is the first AS in the AS path
- These routes are learned directly from ISP1
- This allows us to identify traffic that should use the preferred exit

Step 2: Examine the GW1 IPv4 routing table (before applying local preference).

At this stage of the lab, no Local Preference policy has been applied on GW1. As a result, BGP may select **either ISP1 or ISP2** as the outbound path for external prefixes based on default BGP tie-breaking behavior.

Depending on timing and route processing order, GW1 may choose the path via **ISP2**, even though it is connected over a slower serial link. This behavior is expected and highlights the fact that, without explicit policy, BGP does not consider link speed or design intent when selecting paths.

The output below shows GW1 currently using ISP2 as the next hop for the external prefixes 203.0.112.0/24 through 203.0.115.0/24.

```
GW1# show ip route bgp | begin Gateway
Gateway of last resort is not set

B     192.168.22.0/24 [200/0] via 2.2.2.2, 01:07:14
B     203.0.112.0/24 [20/0] via 209.165.201.18, 00:00:06
B     203.0.113.0/24 [20/0] via 209.165.201.18, 00:00:06
B     203.0.114.0/24 [20/0] via 209.165.201.18, 00:00:06
B     203.0.115.0/24 [20/0] via 209.165.201.18, 00:00:06
```

This reinforces the need for **Local Preference**, which will be configured next to enforce a consistent outbound policy that prefers ISP1 first, ISP2 second, and uses GW2 only as a last resort.

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Step 3: Create a route map to set local preference.

Create a route map on GW1 that uses AS-path matching to set Local Preference values, influencing which ISP link is preferred for outbound traffic from AS 65000.

```
GW1(config)# route-map PREFER-ISP1-ISP2 permit 10
GW1(config-route-map)# match as-path 10
GW1(config-route-map)# set local-preference 200
GW1(config-route-map)# exit
GW1(config)# route-map PREFER-ISP1-ISP2 permit 20
GW1(config-route-map)# set local-preference 150
GW1(config-route-map)# exit
```

Explanation:

- Routes from ISP1 receive Local Preference 200
- All other external routes learned on GW1 receive Local Preference 150
- The default Local Preference (100) is preserved for routes learned via GW2

Step 4: Apply the route map inbound on GW1.

Because **GW1 uses a peer group**, this policy is applied once and inherited by both ISP neighbors. Applying the route map **inbound** (**in**) ensures the Local Preference is set **as routes enter AS 65000**.

```
GW1(config)# router bgp 65000
GW1(config-router)# neighbor ISP-PEERS route-map PREFER-ISP1-ISP2 in
GW1(config-router)# end
```

Step 5: Refresh BGP sessions.

Perform a soft inbound reset to make the new Local Preference values take effect.

```
GW1# clear ip bgp 209.165.201.2 in
GW1# clear ip bgp 209.165.201.18 in
```

These commands perform a **soft inbound reset** of the BGP sessions with ISP1 and ISP2. The BGP adjacencies remain established, but the routes are re-received and reprocessed so the new Local Preference values can be applied without tearing down the sessions.

This does **not** clear the entire BGP table or reset the TCP session; it simply forces the neighbors to resend updates so BGP can recalculate the best paths using the updated policy.

Step 6: Verify local preference.

This output verifies that BGP is selecting best paths based on the configured Local Preference values applied inbound on GW1.

Note: It may take a moment for BGP to reconverge.

```
GW1# show ip bgp
BGP table version is 11, local router ID is 192.168.11.1
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
              r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,
              x best-external, a additional-path, c RIB-compressed,
              t secondary path, L long-lived-stale,
Origin codes: i - IGP, e - EGP, ? - incomplete
RPKI validation codes: V valid, I invalid, N Not found
```

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```
     Network          Next Hop            Metric LocPrf Weight Path
*>   192.168.11.0     0.0.0.0                  0         32768 i
*>i  192.168.22.0     2.2.2.2                  0    100      0 i
*>   203.0.112.0      209.165.201.2                 200      0 65100 65300.3 i
*                     209.165.201.18                150      0 65200 65300.3 i
*>   203.0.113.0      209.165.201.2                 200      0 65100 65300.3 i
*                     209.165.201.18                150      0 65200 65300.3 i
*>   203.0.114.0      209.165.201.2                 200      0 65100 65300.3 i
*                     209.165.201.18                150      0 65200 65300.3 i
*>   203.0.115.0      209.165.201.2                 200      0 65100 65300.3 i
*                     209.165.201.18                150      0 65200 65300.3 i
```

For each external prefix (203.0.112.0/24 through 203.0.115.0/24), GW1 now has **two external paths**:

- One learned directly from **ISP1 (AS 65100)**
- One learned directly from **ISP2 (AS 65200)**
- The Local Preference values clearly reflect the configured policy:
- Routes learned from **ISP1** have a Local Preference of **200**
- Routes learned from **ISP2** have a Local Preference of **150**

Because Local Preference is evaluated **before AS-path length and other attributes**, GW1 correctly selects the ISP1 path as the best path for all external prefixes, as indicated by the ***>** marker.

Routes learned via GW2 do not appear in the BGP table on GW1 because GW1 already has better local exits for those destinations. Because BGP advertises only the best path to iBGP peers by default, alternate paths learned via GW2 are neither selected nor advertised back to GW1.

This confirms that outbound traffic from AS 65000 will:

1) Prefer GW1 → ISP1
2) Fall back to GW1 → ISP2
3) Use GW1 → GW2 → ISP2 only if both GW1 external links fail

Step 7: Re-Examine the GW1 IPv4 routing table (after applying local preference).

Examining the IPv4 routing table on GW1 verifies that the Local Preference policy is now reflected in the IPv4 routing table on GW1, with all external prefixes installed using the preferred exit via ISP1.

```
GW1# show ip route bgp | begin Gateway
Gateway of last resort is not set

B     192.168.22.0/24 [200/0] via 2.2.2.2, 01:28:26
B     203.0.112.0/24 [20/0] via 209.165.201.2, 00:12:20
B     203.0.113.0/24 [20/0] via 209.165.201.2, 00:12:20
B     203.0.114.0/24 [20/0] via 209.165.201.2, 00:12:20
B     203.0.115.0/24 [20/0] via 209.165.201.2, 00:12:20
```

Step 8: Verify GW2 selected the iBGP-learned path from GW1.

Although GW2 has a directly connected eBGP path to ISP2, it selects the iBGP-learned path via GW1 because the **Local Preference value (200) is higher than the default Local Preference (100) of the eBGP path**. **Local Preference** is **evaluated before the eBGP-versus-iBGP check in the BGP decision process**, allowing routing policy defined on GW1 to influence exit selection across the entire AS. This confirms that Local Preference is the correct attribute for enforcing consistent outbound traffic policy within AS 65000.

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Note: In the BGP route selection process, an eBGP path is preferred over iBGP path. This is evaluated after Local Preference.

```
GW2# show ip bgp
BGP table version is 23, local router ID is 192.168.22.1
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
              r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,
              x best-external, a additional-path, c RIB-compressed,
              t secondary path, L long-lived-stale,
Origin codes: i - IGP, e - EGP, ? - incomplete
RPKI validation codes: V valid, I invalid, N Not found

     Network          Next Hop            Metric LocPrf Weight Path
 *>i  192.168.11.0     1.1.1.1                  0    100      0 i
 *>   192.168.22.0     0.0.0.0                  0         32768 i
 *>i  203.0.112.0      1.1.1.1                  0    200      0 65100 4279500803 i
 *                     209.165.201.6                          0 65200 4279500803 i
 *>i  203.0.113.0      1.1.1.1                  0    200      0 65100 4279500803 i
 *                     209.165.201.6                          0 65200 4279500803 i
 *>i  203.0.114.0      1.1.1.1                  0    200      0 65100 4279500803 i
 *                     209.165.201.6                          0 65200 4279500803 i
 *>i  203.0.115.0      1.1.1.1                  0    200      0 65100 4279500803 i
 *                     209.165.201.6                          0 65200 4279500803 i
```

The **i flag** indicates that a route was learned via **iBGP**, while the **absence of i** indicates a route learned directly via **eBGP**.

Step 9: Confirm outbound path selection on GW2.

This output confirms that the BGP best paths selected on GW2 are now installed in the IPv4 routing table, with all external prefixes forwarded via GW1 as dictated by the Local Preference policy.

```
GW2# show ip route bgp | begin Gateway
Gateway of last resort is not set

B     192.168.11.0/24 [200/0] via 1.1.1.1, 01:33:30
B     203.0.112.0/24 [200/0] via 1.1.1.1, 00:17:24
B     203.0.113.0/24 [200/0] via 1.1.1.1, 00:17:24
B     203.0.114.0/24 [200/0] via 1.1.1.1, 00:17:24
B     203.0.115.0/24 [200/0] via 1.1.1.1, 00:17:24
```

Part 6: Configure the MED Attribute

In this part of the lab, you will configure the **Multi-Exit Discriminator (MED)** attribute to influence **inbound traffic** from ISP2 into AS 65000. Unlike Local Preference, which is used internally by AS 65000 to select outbound paths, MED is an attribute that an AS advertises **to a neighboring AS** to suggest which ingress point should be preferred.

In this topology, AS 65000 is multihomed to **ISP2** through two different gateways: **GW1** and **GW2**. The goal is for **ISP2 to prefer the path toward GW2** when sending traffic into AS 65000, rather than entering through GW1. This allows AS 65000 to influence how inbound traffic from ISP2 is distributed across its gateways.

In this part of the lab, you will configure MED attribute on GW1 and GW2 in AS 65000 to influence how ISP2 selects its preferred ingress point to use when sending traffic into AS 65000. Unlike Local Preference, which

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controls outbound traffic inside an AS, MED is used to influence inbound traffic decisions made by a neighboring AS.

It is important to understand that MED has **low priority** in the BGP decision process and is considered **only after attributes such as Weight and Local Preference**—attributes that are controlled by the receiving AS. Because of this, MED is often overridden in real networks and should be viewed as a *hint*, not a guarantee. In practice, MED is most effective when used between directly connected autonomous systems where both sides agree to honor it.

This section demonstrates how MED can be used to influence inbound path selection from ISP2, while also reinforcing why Local Preference remains the primary tool for controlling outbound traffic within AS 65000.

Step 1: Examine the ISP2 BGP routing table (before MED).

Before configuring MED, examine the BGP-learned routes on ISP2. At this stage, ISP2 has no MED information to differentiate the two entry points into AS 65000, so path selection is based on default BGP tie-breaking behavior.

Before MED is configured, ISP2 has no policy-based way to distinguish between GW1 and GW2. As a result, ISP2 may choose either gateway as the preferred next hop for traffic destined to AS 65000, but it will consistently use the same gateway for all prefixes.

```
ISP2# show ip route bgp | begin Gateway
Gateway of last resort is not set

B     192.168.11.0/24 [20/0] via 209.165.201.5, 02:30:18
B     192.168.22.0/24 [20/0] via 209.165.201.5, 02:30:18
B     203.0.112.0/24 [20/0] via 209.165.201.14, 02:30:18
B     203.0.113.0/24 [20/0] via 209.165.201.14, 02:30:18
B     203.0.114.0/24 [20/0] via 209.165.201.14, 02:30:18
B     203.0.115.0/24 [20/0] via 209.165.201.14, 02:30:18
```

Or

```
ISP2# show ip route bgp | begin Gateway
Gateway of last resort is not set

B     192.168.11.0/24 [20/0] via 209.165.201.17, 03:13:16
B     192.168.22.0/24 [20/0] via 209.165.201.17, 03:13:16
B     203.0.112.0/24 [20/0] via 209.165.201.14, 03:12:27
B     203.0.113.0/24 [20/0] via 209.165.201.14, 03:12:27
B     203.0.114.0/24 [20/0] via 209.165.201.14, 03:12:27
B     203.0.115.0/24 [20/0] via 209.165.201.14, 03:12:27
```

Step 2: Configure MED on GW2 (preferred ingress).

GW2 will advertise routes to ISP2 with a **lower MED**, making it the preferred entry point into AS 65000.

```
GW2(config)# route-map SET-MED-ISP2 permit 10
GW2(config-route-map)# set metric 50
GW2(config-route-map)# exit
GW2(config)# router bgp 65000
GW2(config-router)# neighbor 209.165.201.6 route-map SET-MED-ISP2 out
GW2(config-router)# end
```

Explanation:

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- MED is set using the set metric command
- A **lower MED value (50)** makes this path more attractive
- The route map is applied **outbound**, because MED is sent *to* the neighbor
- This configuration tells ISP2 to prefer routes entering AS 65000 via GW2

Step 3: Configure MED on GW1 (backup ingress).

In this step, GW1 advertises routes with a **higher MED value** so that, when ISP2 has a choice of entering AS 65000 via GW1 or GW2, it will **prefer GW2** (lower MED) and treat GW1 as the less-preferred entry point.

Important Note About Peer Groups (Why We Apply the Route Map This Way)

On GW1, ISP1 and ISP2 are members of the **ISP-PEERS** peer group. Because of this, IOS-XE does **not** allow you to apply a route map directly to an individual peer-group member using a **neighbor <ip> route-map ...** command. If you try, IOS-XE rejects it with:

```
% Invalid command for a peer-group member
```

Therefore, the route map must be applied to the **peer group**, which means the MED policy is applied to **both** ISP neighbors on GW1. In a production design, you could use a separate peer group (or more granular policy) to apply MED only toward ISP2, but that added complexity is not required for this lab.

```
GW1(config)# route-map SET-MED-ISP2 permit 10
GW1(config-route-map)# set metric 100
GW1(config-route-map)# exit
GW1(config)# router bgp 65000
GW1(config-router)# neighbor ISP-PEERS route-map SET-MED-ISP2 out
GW1(config-router)# end
```

Explanation:

- GW1 now **sets MED to 100** on routes it advertises to neighbors in the ISP-PEERS peer group.
- This makes GW1 the **less preferred** ingress point compared to GW2 (which will be configured with a lower MED).
- Using explicit MED values on both gateways gives ISP2 a clear basis to compare the two entry points into AS 65000.

Step 4: Refresh BGP sessions with ISP2.

To ensure ISP2 receives the updated MED values, reset the BGP sessions. This clears and reestablishes the BGP sessions so the updated MED values are advertised and processed by ISP2.

```
GW1# clear ip bgp 209.165.201.18
*Jan 25 14:27:34.849: %BGP-3-NOTIFICATION: sent to neighbor 209.165.201.18 6/4
(Administrative Reset) 0 bytes
*Jan 25 14:27:34.850: %BGP-5-ADJCHANGE: neighbor 209.165.201.18 Down User reset
*Jan 25 14:27:34.850: %BGP_SESSION-5-ADJCHANGE: neighbor 209.165.201.18 IPv4 Unicast
topology base removed from session  User reset
*Jan 25 14:27:35.719: %BGP-5-ADJCHANGE: neighbor 209.165.201.18 Up

GW2# clear ip bgp 209.165.201.6
*Jan 25 19:28:21.153: %BGP-3-NOTIFICATION: sent to neighbor 209.165.201.6 6/4
(Administrative Reset) 0 bytes
*Jan 25 19:28:21.164: %BGP-5-ADJCHANGE: neighbor 209.165.201.6 Down User reset
*Jan 25 19:28:21.164: %BGP_SESSION-5-ADJCHANGE: neighbor 209.165.201.6 IPv4 Unicast
topology base removed from session  User reset
```

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```
*Jan 25 19:28:21.411: %BGP-5-ADJCHANGE: neighbor 209.165.201.6 Up
```

Step 5: Verify MED influence on ISP2.

Re-examine the BGP routing table on ISP2. You should now observe that routes destined for AS 65000 consistently prefer the path via **GW2**, reflecting the lower MED value advertised by GW2.

```
ISP2# show ip bgp
BGP table version is 15, local router ID is 4.4.4.4
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
              r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,
              x best-external, a additional-path, c RIB-compressed,
              t secondary path, L long-lived-stale,
Origin codes: i - IGP, e - EGP, ? - incomplete
RPKI validation codes: V valid, I invalid, N Not found

     Network          Next Hop            Metric LocPrf Weight Path
*>   192.168.11.0     209.165.201.5            50             0 65000 i
*                     209.165.201.17          100             0 65000 i
*>   192.168.22.0     209.165.201.5            50             0 65000 i
*                     209.165.201.17          100             0 65000 i
*>   203.0.112.0      209.165.201.14             0             0 4279500803 i
*>   203.0.113.0      209.165.201.14             0             0 4279500803 i
*>   203.0.114.0      209.165.201.14             0             0 4279500803 i
*>   203.0.115.0      209.165.201.14             0             0 4279500803 i
```

From the output:

- For prefixes **192.168.11.0/24** and **192.168.22.0/24**, ISP2 now sees **multiple paths** into AS 65000:
  - Via **GW2 (209.165.201.5)** with **MED 50**
  - Via **GW1 (209.165.201.17)** with **MED 100**
- Because **lower MED values are preferred**, ISP2 selects the paths via **GW2**, as indicated by the *> best-path marker.
- This demonstrates that ISP2 is honoring the MED values advertised by AS 65000 and consistently prefers **GW2 as the ingress point**.

The external prefixes (203.0.112.0/24–203.0.115.0/24) continue to use ISP3 as expected and are unaffected by MED, since MED only applies when comparing multiple paths **from the same neighboring AS**.

Confirmation

- MED is influencing inbound traffic decisions made by ISP2
- Lower MED values are preferred
- MED is only compared for routes received from the same neighboring AS (AS 65000)
- MED is evaluated late in the BGP decision process and can be overridden by provider policy
- Explicitly configuring MED on both gateways produces predictable and realistic inbound behavior

Why MED Is Often Ignored in Real Provider Networks?

Although MED can influence inbound traffic decisions, it is **one of the weakest and least reliable BGP attributes** in real-world service provider environments.

In practice, many providers **do not honor MED values received from customers** for several reasons:

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- **Customer-controlled**: MED is set by the *advertising AS*, not the receiving AS. Providers are often unwilling to let customers influence their internal routing decisions.
- **Evaluated late**: MED is considered **after** Weight, Local Preference, and AS-path length in the BGP decision process. Providers typically use **Local Preference** to enforce their own policies, which overrides MED.
- **Limited scope**: MED is only compared between routes received from the **same neighboring AS**, which limits its usefulness in complex, multi-provider topologies.
- **Operational complexity**: Honoring MED at scale can complicate traffic engineering and troubleshooting for large networks.

As a result, many service providers either **ignore MED entirely**, normalize it, or override it with internal policy.

Why We Still Use MED in This Lab

- Despite its limitations, MED remains an important concept to understand because:
- It demonstrates how an AS can **suggest** a preferred ingress point
- It reinforces the distinction between **inbound vs. outbound** traffic engineering
- It is still used in some controlled environments (such as between cooperating networks or within a single administrative domain)

In this lab, ISP2 honors MED to clearly illustrate how the attribute works and how it compares to stronger attributes like Local Preference.

Part 7: Configure ISP1 to Advertise a Default Route in BGP

In many real-world deployments, an ISP does not advertise a full Internet routing table to customer networks. Instead, the ISP may advertise a **default route (0.0.0.0/0)**, a **partial set of Internet routes** (often learned from selected peers), or a combination of both. A default route effectively tells the customer, *"Send all traffic for unknown destinations to me, and I will take care of routing it."*

This approach simplifies the customer's routing table, reduces memory and CPU requirements, and clearly establishes the ISP as the upstream exit point for Internet-bound traffic. It is especially common when the customer is multihomed but still wants a preferred primary provider for general outbound traffic.

From the customer's perspective, receiving a default route allows internal routers to reach external networks without requiring knowledge of every possible destination prefix. From the ISP's perspective, advertising a default route is a controlled way to provide connectivity while maintaining policy and scalability.

In this part of the lab, **ISP1 acts as an upstream provider** for AS 65000. Rather than advertising a full routing table, ISP1 will advertise a default route *only* to GW1. This mirrors a realistic ISP–customer relationship and gives AS 65000 a clear, controlled exit point for Internet-bound traffic.

Ways to Advertise a Default Route in BGP

There are several ways an ISP can advertise a default route using BGP. While all achieve the same basic outcome, they differ significantly in scope, control, and realism.

network 0.0.0.0/0

This method injects a default route into BGP *only if* a default route already exists in the routing table. Once injected, it is advertised to all BGP neighbors unless explicitly filtered.

- Requires an existing default route in the RIB
- Advertised to all neighbors by default
- Common in internal or controlled environments

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redistribute static or redistribute <protocol>

This method redistributes routes from another routing source into BGP, including a default route if one exists.

- Can unintentionally inject many routes
- Requires careful route-map filtering
- Less precise for teaching or customer-specific policy

default-information originate

This command causes the router to *artificially generate* a default route in BGP, regardless of whether one exists in the routing table.

- Advertised to all BGP neighbors
- Useful when the router is clearly an upstream exit
- Still broad in scope

neighbor default-originate (Most Targeted)

This command also generates a default route artificially, but advertises it **only to a specific neighbor**.

- No requirement for an existing default route
- Advertised only to selected peers
- Very common in ISP–customer designs
- Ideal for teaching precise policy control

Why We Use neighbor default-originate in This Lab

Using neighbor **default-originate** allows us to:

- Clearly demonstrate how default routes enter BGP
- Avoid unnecessary complexity
- Maintain tight control over which neighbors receive the default
- Show how the default route is propagated via iBGP from GW1 to GW2

This approach aligns perfectly with the design goals of the lab and reinforces best practices used in production networks.

Step 1: Advertise a default route from ISP1 to GW1.

ISP1 will use the **neighbor default-originate** command to advertise an artificially generated default route **only to GW1**. This method is commonly used by providers because it does not require a default route to exist in the ISP's routing table and allows precise control over which neighbors receive the default.

```
ISP1(config)# router bgp 65100
ISP1(config-router)# neighbor 209.165.201.1 default-originate
ISP1(config-router)# end
```

Explanation

- neighbor 209.165.201.1 default-originate causes ISP1 to generate and advertise 0.0.0.0/0 to GW1
- The default route is not advertised globally, only to the specified neighbor
- This reflects a typical ISP policy where default routing is provided selectively to customers

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Note: It may take a few moments for the default route to appear on GW1 and propagate to GW2 via iBGP.

Step 2: Verify the default route on GW1.

Examine the BGP table on GW1 to confirm that the default route has been learned from ISP1.

Note: It may take a few minutes for BGP to reconverge.

```
GW1# show ip bgp
BGP table version is 24, local router ID is 192.168.11.1
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
              r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,
              x best-external, a additional-path, c RIB-compressed,
              t secondary path, L long-lived-stale,
Origin codes: i - IGP, e - EGP, ? - incomplete
RPKI validation codes: V valid, I invalid, N Not found

     Network          Next Hop            Metric LocPrf Weight Path
*>   0.0.0.0          209.165.201.2                200      0 65100 i
<output omitted>
```

Explanation

- The ***>** indicates this is the selected **best path**
- The **default route** is treated like **any other BGP-learned prefix**
- **Local Preference (200)** reflects the **policy previously configured to prefer ISP1**

Step 3: Verify the default route in the IPv4 routing table on GW1.

Confirm that the BGP default route has been installed in the IP routing table.

```
GW1# show ip route bgp | begin Gateway
Gateway of last resort is 209.165.201.2 to network 0.0.0.0

B*    0.0.0.0/0 [20/0] via 209.165.201.2, 00:05:17
B     192.168.22.0/24 [200/0] via 2.2.2.2, 04:06:37
B     203.0.112.0/24 [20/0] via 209.165.201.2, 02:50:31
B     203.0.113.0/24 [20/0] via 209.165.201.2, 02:50:31
B     203.0.114.0/24 [20/0] via 209.165.201.2, 02:50:31
B     203.0.115.0/24 [20/0] via 209.165.201.2, 02:50:31
```

Step 4: Verify default route propagation to GW2.

Because GW1 and GW2 exchange routes using iBGP, GW1 advertises the default route internally to GW2.

```
GW2# show ip bgp
BGP table version is 24, local router ID is 192.168.22.1
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
              r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,
               x best-external, a additional-path, c RIB-compressed,
              t secondary path, L long-lived-stale,
Origin codes: i - IGP, e - EGP, ? - incomplete
RPKI validation codes: V valid, I invalid, N Not found

     Network          Next Hop            Metric LocPrf Weight Path
```

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```
*>i  0.0.0.0          1.1.1.1                  0    200      0 65100 i
<output omitted>
```

Step 5: Verify the default route in the IPv4 routing table on GW2.

Examine.

```
GW2# show ip route bgp | begin Gateway
Gateway of last resort is 1.1.1.1 to network 0.0.0.0

B*    0.0.0.0/0 [200/0] via 1.1.1.1, 00:13:34
B     192.168.11.0/24 [200/0] via 1.1.1.1, 04:14:54
B     203.0.112.0/24 [200/0] via 1.1.1.1, 02:58:48
B     203.0.113.0/24 [200/0] via 1.1.1.1, 02:58:48
B     203.0.114.0/24 [200/0] via 1.1.1.1, 02:58:48
B     203.0.115.0/24 [200/0] via 1.1.1.1, 02:58:48
```

Step 6: Verify forwarding using traceroute from GW2.

Finally, verify that GW2 can reach a destination **not explicitly advertised** via BGP by relying on the default route. The destination **209.165.201.21** is a loopback interface on ISP1 that is not advertised as a specific BGP prefix. This traceroute confirms that GW2 forwards unknown destinations to GW1 using the iBGP-learned default route, and GW1 then forwards the traffic to ISP1, which provides Internet reachability for AS 65000.

```
GW2# traceroute 209.165.201.21 source Loopback1
Type escape sequence to abort.
Tracing the route to 209.165.201.21
VRF info: (vrf in name/id, vrf out name/id)
  1 172.16.12.1 [AS 65100] 89 msec 1 msec 0 msec
  2 209.165.201.2 [AS 65100] 13 msec *  1 msec
```

Part 8: Configure ISP3 to Advertise a Summary (Aggregate) of its Prefixes

In earlier parts of this lab, ISP3 (AS **65300.3**) advertised four individual IPv4 prefixes into BGP:

- 203.0.112.0/24
- 203.0.113.0/24
- 203.0.114.0/24
- 203.0.115.0/24

While advertising individual prefixes provides precise routing information, it does not scale well. On the public Internet, service providers typically **aggregate related prefixes into a single summary route** before advertising them upstream or to customers.

This process—called **BGP aggregation**—reduces:

- The size of the global BGP routing table
- Memory and CPU usage on routers
- The impact of route flapping

In this part of the lab, ISP3 will summarize its four /24 loopback networks into a single **203.0.112.0/22** route and advertise that summary to the rest of the network.

What Is BGP Aggregation?

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BGP aggregation allows a router to:

- Inject a **summary route** into the BGP table when at least one more-specific route within that range exists
- Optionally **suppress** the more-specific routes from being advertised

Important characteristics:

- The more-specific routes **remain in the local BGP table**
- Suppressed routes are **not advertised** to BGP neighbors
- The summary remains advertised as long as **at least one component route exists**

This makes aggregation both **efficient** and **stable**.

Step 1: Configure a BGP aggregate on ISP3 (without suppression).

In this step, ISP3 will generate a summary route for its four /24 prefixes **without suppressing** the more-specific routes. This allows you to observe both the aggregate and the individual routes being advertised.

```
ISP3(config)# router bgp 65300.3
ISP3(config-router)# aggregate-address 203.0.112.0 255.255.252.0
ISP3(config-router)# end
```

Explanation

- The aggregate **203.0.112.0/22** covers:
  - 203.0.112.0/24
  - 203.0.113.0/24
  - 203.0.114.0/24
  - 203.0.115.0/24
- Because these routes already exist in the BGP table, BGP injects the aggregate route
- At this stage, **both the summary and the more-specific routes are advertised**

Step 2: Verify the aggregate route on GW1.

Examine the BGP table on GW1 to confirm that aggregation alone does **not** reduce routing information unless suppression is explicitly enabled.

Note: It may take a moment for BGP to reconverge.

```
GW1# show ip bgp
BGP table version is 25, local router ID is 192.168.11.1
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
              r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,
              x best-external, a additional-path, c RIB-compressed,
              t secondary path, L long-lived-stale,
Origin codes: i - IGP, e - EGP, ? - incomplete
RPKI validation codes: V valid, I invalid, N Not found

     Network          Next Hop            Metric LocPrf Weight Path
 *>   0.0.0.0          209.165.201.2                200      0 65100 i
 *>   192.168.11.0     0.0.0.0                   0         32768 i
 *>i  192.168.22.0     2.2.2.2                   0    100      0 i
 *    203.0.112.0      209.165.201.18               150      0 65200 65300.3 i
```

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```
 *>                   209.165.201.2                200      0 65100 65300.3 i
 *    203.0.112.0/22  209.165.201.18               150      0 65200 65300.3 i
 *>                   209.165.201.2                200      0 65100 65300.3 i
 *    203.0.113.0     209.165.201.18               150      0 65200 65300.3 i
 *>                   209.165.201.2                200      0 65100 65300.3 i
 *    203.0.114.0     209.165.201.18               150      0 65200 65300.3 i
 *>                   209.165.201.2                200      0 65100 65300.3 i
 *    203.0.115.0     209.165.201.18               150      0 65200 65300.3 i
 *>                   209.165.201.2                200      0 65100 65300.3 i
```

Expected Observations

- The aggregate route **203.0.112.0/22** appears in the BGP table
- The individual /24 routes are **still present**
- Multiple paths may exist (via ISP1 and ISP2)

Step 3: Verify Summary and Specific Routes in the IPv4 routing table on GW1.

Although ISP3 is advertising an aggregate route (203.0.112.0/22), GW1 has also learned the individual /24 routes that fall within that summary. All of these routes are installed in the routing table, as shown above. However, **IP forwarding always uses longest-prefix match**, so traffic destined for 203.0.112.0/24–203.0.115.0/24 is forwarded using the more specific /24 routes, not the /22 summary.

The /22 route is present in the routing table and is used only when no more specific prefix match exists or is available, since IP forwarding always prefers the longest prefix match. This demonstrates why BGP aggregation without the summary-only keyword does not change forwarding behavior—it merely adds an additional, less specific route.

```
GW1# show ip route bgp | begin Gateway
Gateway of last resort is 209.165.201.2 to network 0.0.0.0

B*    0.0.0.0/0 [20/0] via 209.165.201.2, 00:46:00
B     192.168.22.0/24 [200/0] via 2.2.2.2, 04:47:20
B     203.0.112.0/22 [20/0] via 209.165.201.2, 00:03:20
B     203.0.112.0/24 [20/0] via 209.165.201.2, 03:31:14
B     203.0.113.0/24 [20/0] via 209.165.201.2, 03:31:14
B     203.0.114.0/24 [20/0] via 209.165.201.2, 03:31:14
B     203.0.115.0/24 [20/0] via 209.165.201.2, 03:31:14
```

Step 4: Suppress the more-specific routes (summary-only).

In most real networks, advertising both the summary and the specifics defeats the purpose of aggregation. In this step, ISP3 will suppress the individual /24 routes and advertise **only the summary**.

```
ISP3(config)# router bgp 65300.3
ISP3(config-router)# no aggregate-address 203.0.112.0 255.255.252.0
ISP3(config-router)# aggregate-address 203.0.112.0 255.255.252.0 summary-only
ISP3(config-router)# end
```

Explanation

- The summary-only keyword suppresses the component routes
- The more specific routes remain in ISP3's BGP table but are **not advertised**
  - Only the aggregate **203.0.112.0/22** is sent to neighbors

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Step 5: Verify aggregation on ISP3.

Examine the BGP table on ISP3 to confirm that ISP3 is performing true route aggregation.

Note: It may take a moment for BGP to reconverge.

```
ISP3# show ip bgp
BGP table version is 14, local router ID is 203.0.115.1
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
              r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,
              x best-external, a additional-path, c RIB-compressed,
              t secondary path, L long-lived-stale,
Origin codes: i - IGP, e - EGP, ? - incomplete
RPKI validation codes: V valid, I invalid, N Not found

     Network          Next Hop            Metric LocPrf Weight Path
 *    192.168.11.0     209.165.201.13                         0 65200 65000 i
 *>                    209.165.201.9                          0 65100 65000 i
 *    192.168.22.0     209.165.201.9                          0 65100 65000 i
 *>                    209.165.201.13                         0 65200 65000 i
 s>   203.0.112.0      0.0.0.0                  0         32768 i
 *>   203.0.112.0/22   0.0.0.0                            32768 i
 s>   203.0.113.0      0.0.0.0                  0         32768 i
 s>   203.0.114.0      0.0.0.0                  0         32768 i
 s>   203.0.115.0      0.0.0.0                  0         32768 i
```

Expected Observations

- The aggregate route **203.0.112.0/22** appears as valid and best
- The /24 routes are marked with **s (suppressed)**
- Suppressed routes are not advertised to any peers

Step 6: Verify that GW1 receives only the summary route.

Examine the BGP table on GW1 to demonstrate the **downstream benefit** of aggregation.

```
GW1# show ip bgp
BGP table version is 33, local router ID is 192.168.11.1
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
              r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,
              x best-external, a additional-path, c RIB-compressed,
              t secondary path, L long-lived-stale,
Origin codes: i - IGP, e - EGP, ? - incomplete
RPKI validation codes: V valid, I invalid, N Not found

     Network          Next Hop            Metric LocPrf Weight Path
 *>   0.0.0.0          209.165.201.2                200      0 65100 i
 *>   192.168.11.0     0.0.0.0                  0         32768 i
 *>i  192.168.22.0     2.2.2.2                  0    100      0 i
 *>   203.0.112.0/22   209.165.201.2                200      0 65100 65300.3 i
 *                     209.165.201.18               150      0 65200 65300.3 i
```

Expected Observations

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- Only **203.0.112.0/22** is present
- The individual **/24 routes no longer appear**
- GW1 sees a much **smaller routing table**

Step 7: Verify suppression of specific routes in GW1 and GW2 IPv4 routing tables.

After enabling aggregation with **summary-only** on ISP3, the more specific /24 routes (203.0.112.0/24 – 203.0.115.0/24) are no longer advertised beyond ISP3. As a result, both GW1 and GW2 now receive and install **only the aggregated route 203.0.112.0/22** in their BGP routing tables.

On **GW1**, the /22 route is learned from ISP1 and installed with the preferred next hop of **209.165.201.2**, consistent with the previously configured Local Preference policy. On **GW2**, the same /22 route is learned via iBGP from GW1 and installed with next hop **1.1.1.1**, making GW1 the exit point toward ISP1 for traffic destined to the summarized prefixes.

Because no more specific routes exist, the /22 aggregate is now the **longest prefix match** and is used for forwarding on both routers. This confirms that route suppression is working as intended and that the summary route provides reachability while reducing routing table size and update churn.

```
GW1# show ip route bgp | begin Gateway
Gateway of last resort is 209.165.201.2 to network 0.0.0.0

B*    0.0.0.0/0 [20/0] via 209.165.201.2, 01:03:30
B     192.168.22.0/24 [200/0] via 2.2.2.2, 05:04:50
B     203.0.112.0/22 [20/0] via 209.165.201.2, 00:04:22

GW2# show ip route bgp | begin Gateway
Gateway of last resort is 1.1.1.1 to network 0.0.0.0

B*    0.0.0.0/0 [200/0] via 1.1.1.1, 01:03:58
B     192.168.11.0/24 [200/0] via 1.1.1.1, 05:05:18
B     203.0.112.0/22 [200/0] via 1.1.1.1, 00:04:50
```

Step 8: Verify end-to-end connectivity using the summary.

Even though the /24 routes are suppressed, traffic should still reach individual destinations within the summary.

```
GW2# ping 203.0.114.1 source Loopback1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 203.0.114.1, timeout is 2 seconds:
Packet sent with a source address of 192.168.22.1
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/3 ms
```

Why This Matters (Real-World Perspective).

BGP aggregation is critical for Internet scalability:

Benefits

- Smaller BGP tables
- Reduced memory and CPU usage
- Fewer routing updates

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- Improved stability during partial failures

Trade-offs

- Loss of routing granularity
- Potentially suboptimal path selection when multiple paths exist
- Less visibility into individual subnets

This is why aggregation is used carefully and typically **at network boundaries**, not within highly meshed core networks.

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